Product	Metric	Previous period	Current period
Rice crop	Yield per hectare	4.0 tonnes	3.5 tonnes
COVID-19 test kits	Monthly demand	10,000 kits	14,000 kits
Electric vehicles	Monthly sales	500 units	650 units
Solar panels	Energy generated per day	1,000 kWh	1,200 kWh
Cement	Production per month	20,000 tonnes	18,000 tonnes

-Descriptive analysis — What happened?

1. Rice crop

   Yield decreased from 4.0 to 3.5 tonnes per hectare, a decline of 12.5%.

2. COVID-19 test kits

   Monthly demand increased from 10,000 to 14,000 kits, a rise of 40%.

3. Electric vehicles

   Monthly sales increased from 500 to 650 vehicles, a rise of 30%.

4. Solar panels

   Daily energy generation increased from 1,000 to 1,200 kWh, a rise of 20%.

5. Cement

   Monthly production decreased from 20,000 to 18,000 tonnes, a decline of 10%.

-Diagnostic analysis — Why did it happen?

This analysis investigates possible reasons for the changes.

Product	Possible reasons to investigate
Rice crop	Insufficient rainfall, pests, soil quality or fertilizer issues
COVID-19 test kits	Increased infections, institutional orders or seasonal demand
Electric vehicles	Government incentives, new models or improved charging infrastructure
Solar panels	Higher sunlight availability, reduced downtime or improved maintenance
Cement	Machinery breakdowns, raw material shortages or reduced demand

-Predictive analysis — What might happen next?

1. Rice Crop: If unfavorable weather conditions and insufficient irrigation continue, rice production may decrease further in the coming seasons. However, improved rainfall and farming practices could help increase crop yield.

2. COVID-19 Test Kits: Demand for COVID-19 test kits may increase if infections rise or new outbreaks occur. If infection rates remain low, demand may decrease.

3. Electric Vehicles: Demand for electric vehicles is likely to grow as consumers become more interested in sustainable transportation, charging infrastructure improves, and battery technology advances.

4. Solar Panels: Solar energy generation may increase as more solar installations are developed and panel efficiency improves. However, weather conditions, dust accumulation, and equipment failures could affect energy output.

5. Cement: Cement production may increase if construction and infrastructure projects expand. However, economic slowdowns, reduced construction activity, or raw material shortages could reduce future production.

-Prescriptive analysis — What should we do?

Product	**Recommended action**
Rice crop	Test soil, optimize irrigation and investigate pest damage
COVID-19 test kits	Adjust production and inventory to meet verified demand
Electric vehicles	Plan production capacity and secure battery supplies
Solar panels	Maintain equipment and investigate factors affecting energy output
Cement	Perform preventive maintenance and investigate production bottlenecks